

Long COVID whole-blood and PBMC programs after preregistration: a cross-assay oxidative-phosphorylation signal, two transparent negatives on ME/CFS separation, and a full gate audit

MEGA-PROGRAM-27, Long COVID lane (builder branch builder-25-ic)

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Abstract

Long COVID (post-acute sequelae of SARS-CoV-2 infection, PASC) still lacks any molecular test; its closest clinical neighbour, myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), shares both symptoms and putative biology, and the two are routinely conflated in cohort design. This report runs a disciplined, fully preregistered program on four public cohorts and keeps every negative on the record. First, three fixed-list tests derived from published six-gene panels fail their strict rules: the GSE226260-derived list shows 5/6 expected signs in its own source cohort ($p = .0081$, no Bonferroni pass), fails outright on a NanoString PBMC cohort, and a naive-CD4 cohort cannot measure CXCL8 at all. Second, the preregistered P51 arc trains nine Hallmark-derived modules on GSE226260 whole blood (17 PASC vs 20 recovered) and selects exactly one: oxidative phosphorylation, down-regulated in PASC (permutation $p = .0432$, leave-one-patient-out stable 37/37). Frozen evaluation on the untouched GSE275334 PBMC NanoString cohort shows the oxphos-down signature separates Long COVID from healthy controls cross-assay (E1: $p = .0038$, Hedges $g = 0.958$) but does not separate Long COVID from ME/CFS (E2: $p = .485$, $g = 0.259$); the published six-gene comparator also fails the locked $|g| > 0.8$ bar on that contrast ($g = 0.776$). The locked win rule is not met and is recorded as such. Third, the preregistered P52 arc - designed after an advisory design review and correcting the first arc's anti-conservative one-sided selection to a two-sided, platform-stratified rule - attempts to train an ME/CFS-contrast signature on post-infectious ME/CFS PBMC (GSE251872; 12 cases, 15 controls): no module passes (best two-sided $p = .194$), and the arc ends at the training stage, by preregistration, without touching the evaluation cohort. Fourth, we build the lane's gate evidence honestly: 178 accession-backed record units (record units, not patients or studies), 44 genuinely used external scientific services (42 under the conservative recount), and this manuscript toward the 50-substantive-page gate, whose status is stated exactly wherever it stands. No clinical test, named discovery, or benchmark break is claimed.

Contents

1	Scope, verdict and reproducibility	3
2	Disease background and the ME/CFS adjacency problem	3
3	Data acquisition and provenance	3
3.1	Datasets and roles	3
3.2	Verification	4
4	Pre-arc fixed-list tests: three negatives that forced preregistration	4

5	Preregistered arc P51: the oxidative-phosphorylation signal	4
5.1	Frozen design	4
5.2	Training result	4
5.3	Evaluation result and the honest ledger	5
5.4	The anti-conservatism note, carried forward	5
6	Preregistered arc P52: the ME/CFS-contrast attempt ends at training	5
6.1	Why a second arc, and what changed	5
6.2	Cohort choice, made and justified before any analysis	5
6.3	Advisory design review, applied before the freeze	5
6.4	Frozen protocol and the single execution	6
6.5	Result: a clean training-stage negative	6
7	External-service annotation landscape	6
7.1	Purpose and accounting	6
7.2	What the landscape actually says	6
7.3	Independent resonance	7
8	Gate audit, stated exactly	7
9	Limitations	7
10	Reproducibility appendix	8
11	Statistical methods appendix	8
11.1	Module scoring	8
11.2	Effect sizes and permutation tests	8
11.3	Eligibility bars, locked a priori	9
11.4	Complete module-level results	9

1 Scope, verdict and reproducibility

Scope. This manuscript is the disease-lane report for the Long COVID project of MEGA-PROGRAM-27. It covers: (i) provenance and verification of every dataset the lane has used or credited; (ii) the pre-arc fixed-list tests that motivated preregistration; (iii) the two preregistered analysis arcs, P51 and P52, with their frozen protocols, single executions, and untuned outcomes; (iv) the external-service annotation landscape that contextualises the preregistered gene sets; and (v) the lane’s gate audit against the program’s 120-record, 40-service and 50-substantive-page floors.

Verdict, stated up front. The lane holds one preregistered positive and two preregistered negatives. The positive: an oxidative-phosphorylation down-regulation signature, selected under a frozen protocol in whole blood, transports across assay platform and cell mixture to separate Long COVID from healthy controls in PBMC (E1, $p = .0038$). The negatives: that signature does not distinguish Long COVID from ME/CFS (E2, $p = .485$), and a second, better-powered-design arc aimed directly at the ME/CFS contrast found no trainable module signal at all (best two-sided $p = .194$). We claim the positive exactly as far as the preregistration supports: a reproducible chronic-state signal against healthy controls, not a disease-specific discriminator and not a diagnostic. The ME/CFS separation question - the lane’s declared differentiator in its novelty plan - is, after two disciplined attempts, unresolved: the first arc’s signatures fail it, and the second arc shows the failure is not fixed by switching training cohorts at this sample size.

Reproducibility. Every dataset is stored in-repo under `data/geo/` with SHA-256 locks asserted at the top of every analysis script. Every gated test has its preregistration committed and pushed before its first execution (`projects/long_covid/prereg/`). Every analysis script runs end-to-end from the locked data with a fixed seed and writes a JSON result committed alongside. Advisory consultations (judge mode) are preserved verbatim with conversation URLs in `redirect/`. One implementation bug occurred (gzip handling in the P52 trainer); it failed before producing any result, is disclosed in the redirect log, and the fixed script’s output is the single training execution the preregistration allows.

2 Disease background and the ME/CFS adjacency problem

Long COVID denotes persistent symptoms following SARS-CoV-2 infection that cannot be explained by an alternative diagnosis; fatigue, post-exertional malaise, cognitive dysfunction and autonomic features dominate. Its case definition is clinical and exclusionary, which makes a molecular discriminator valuable in principle and dangerous in practice: any blood signal that merely tracks chronic illness will appear to “work” against healthy controls while adding nothing over clinical assessment. The strongest version of the problem is ME/CFS. ME/CFS is likewise a post-infectious, exclusion-diagnosed syndrome with overlapping symptomatology; a nontrivial fraction of Long COVID patients meet ME/CFS criteria. A biomarker claim for Long COVID that has never been tested against ME/CFS controls is, at best, a claim about chronic unwellness.

This is why the lane’s novelty plan names the ME/CFS contrast as the actual contribution to attempt, and why both preregistered arcs are built around it: P51 asks whether a discovered signature separates Long COVID from ME/CFS, not merely from health; P52 asks whether a signature trained on ME/CFS itself separates the two. The honest scientific prior, which the advisory review of the P52 design stated plainly, is that these conditions may be molecularly continuous in blood; the program’s job is to test that falsifiably, not to assume separability.

3 Data acquisition and provenance

3.1 Datasets and roles

Four GEO series are locked in-repo. **GSE226260** (whole-blood RNA-seq; source study Nature Immunology 2025) contributes the P51 training cohort: 17 PASC vs 20 recovered donors, one sample per person

selected under the lane’s sample-map rules from 103 libraries (`results/longcovid_p36_samples.csv`); the same series is the source of the published six-gene panel (JAK1, CXCL8, BCL2L1, OSM, MAP3K8, STAT3) used throughout as the named comparator. **GSE275334** (PBMC, 635-gene NanoString panel) contributes the locked evaluation cohort: 23 Long COVID, 9 ME/CFS, 15 healthy controls (`results/longcovid_p37_samples.csv`) - to our knowledge the only public cohort carrying Long COVID and ME/CFS arms on one platform, which is why it is reserved for evaluation and never trained on. **GSE334857** (naive CD4 T-cells) contributes the p38 coverage test. **GSE251872** (Walitt deep-phenotyping post-infectious ME/CFS, baseline PBMC RNA-seq) contributes the P52 training cohort: 12 cases vs 15 healthy volunteers across two sequencing platforms (3/7 and 9/8 case/control splits), a confound the P52 preregistration handles by within-platform z-scoring and platform-stratified permutation.

3.2 Verification

All files carry SHA-256 digests asserted in-script before any processing; sample-to-group assignments are taken from series-matrix titles and characteristics, not from secondary citations. The GSE251872 group split (12/15) and platform split were re-derived from the two platform-specific series matrices and are re-asserted in the training script, which fails closed if the composition ever changes.

4 Pre-arc fixed-list tests: three negatives that forced preregistration

Before any preregistered arc, the lane tested the obvious cheap hypotheses - published fixed gene lists - under strict all-gene rules (`results/longcovid_p36_result.json`, p37, p38). The GSE226260 six-gene list, evaluated in its own source cohort, shows 5/6 expected signs ($p = .0081$) but no gene passes a Bonferroni bar: a weak, post-selected signal exactly where it was discovered. The same list fails its strict rule outright on GSE275334. The naive-CD4 cohort cannot measure CXCL8 (the list’s anchor gene) above detection, so the third test is uninterpretable by construction and recorded as a coverage failure, not a result. These negatives are why the lane moved to frozen-protocol module arcs: post-selected six-gene lists do not survive contact with independent cohorts, and any discovered signature must be selected where it is trained and judged where it has never been seen.

5 Preregistered arc P51: the oxidative-phosphorylation signal

5.1 Frozen design

The P51 preregistration (committed at 42deadfa before any run) fixed: nine Hallmark-derived candidate modules L1–L9 (`prereg/lc_p51_module_genesets.json`); training on GSE226260 only; module eligibility bars (≥ 20 genes measurable in training, ≥ 10 on the 635-gene evaluation panel, locked a priori because the panel is small); selection by Cohen’s D plus permutation testing with leave-one-patient-out sign stability; and a frozen evaluation on GSE275334 comprising E1 (signature vs healthy, one-sided in the trained direction, pass $p < .025$), E2 (signature vs ME/CFS, two-sided $p < .05$ and $|g| > 0.8$), the same two tests for the six-gene comparator on the same subjects with the same machinery, and a win rule requiring E1 and E2 to pass and the signature’s E2 effect to exceed the comparator’s.

5.2 Training result

One module of nine was selected: L7, oxidative phosphorylation, down-regulated in PASC vs recovered ($D = -0.356$, permutation $p = .0432$, sign stable in all 37 leave-one-out folds; `results/lc_p51_train_result.json`). The other eight modules show permutation p between .10 and .29 - no second signal. Twelve of the module’s genes are on the evaluation panel, clearing the locked ≥ 10 bar.

5.3 Evaluation result and the honest ledger

On GSE275334 (results/lc_p51_eval_result.json): **E1 passes** - the oxphos-down signature separates Long COVID from healthy controls cross-assay ($D = 0.343$, $p = .0038$, $g = 0.958$). **E2 fails** - it does not separate Long COVID from ME/CFS ($D = 0.113$, $p = .485$, $g = 0.259$). The six-gene comparator is stronger on both tests ($p = .0018$, $g = 1.051$ against healthy; $p = .0473$, $g = 0.776$ against ME/CFS) yet itself fails the locked $|g| > 0.8$ bar on the ME/CFS contrast. The win rule is not met; the arc is recorded as a ledger outcome, untuned.

5.4 The anti-conservatism note, carried forward

The P51 training permutation was implemented one-sided in each module's observed direction; because direction was not fixed a priori, this is anti-conservative relative to a truly prespecified test. The flaw is disclosed in the result commit, the training result was not re-run (no tuning into a different answer), and the correction was mandated into the P52 preregistration: two-sided selection, stated below.

6 Preregistered arc P52: the ME/CFS-contrast attempt ends at training

6.1 Why a second arc, and what changed

After P51, the lane's open question was no longer whether a blood signal separates Long COVID from health - it does - but whether anything we can train separates Long COVID from ME/CFS. The program's direction was to aim a second arc directly at that contrast, incorporating the P51 self-correction (no observed-direction one-sided testing). The design problem is data: the P51 training cohort has no ME/CFS arm, so an ME/CFS-contrast signature must be trained on an ME/CFS cohort and evaluated for transport into the locked GSE275334 evaluation set.

6.2 Cohort choice, made and justified before any analysis

Four public ME/CFS expression cohorts were reconned from GEO metadata alone (titles, summaries, sample compositions - no expression values): GSE293840 (168 samples, but cell-free plasma RNA - an analyte that cannot transport to a cellular NanoString panel), GSE227375 (35 patients, 34 controls, whole blood, exercise-challenge design), GSE128078 (14 patients, 11 controls, longitudinal exercise), and GSE251872 (the Walitt deep-phenotyping post-infectious ME/CFS study, baseline PBMC RNA-seq). GSE251872 was chosen because it is the only post-infectious ME/CFS cohort (the subtype etiologically closest to Long COVID) and the only PBMC cohort (tissue-matched to the evaluation panel). Its cost is power: 12 cases and 15 controls, split across two platforms (3/7 and 9/8). The preregistration accepted that cost explicitly and handled the platform confound by within-platform z-scoring and platform-stratified permutation.

6.3 Advisory design review, applied before the freeze

A judge-mode advisory review of the draft design (verbatim record and conversation URL in [redirect/lc_p52_judge-co](#)) found no fatal flaw and three serious risks: the tiny training n, residual platform-identification risk, and RNA-seq-to-NanoString transport validity. It judged the planned beat-the-comparator win gate stacked (the comparator panel was post-selected by its source paper) and recommended a preregistered platform sensitivity check. Three adaptations were applied to the preregistration before it froze: the comparator leg was demoted from hard gate to reported comparison with a bootstrap interval; a leave-one-platform-out direction check was added as a locked, report-only sensitivity; and module gene composition was fixed at selection with the evaluation using only the pre-declared panel intersection. The review's framing is adopted verbatim in spirit: this arc is a high-integrity falsification test; its failure indicts this signature strategy, not the possibility of molecular separation.

6.4 Frozen protocol and the single execution

The P52 preregistration (committed at f81bb92d before any run) fixed: the same nine L1–L9 modules; training on GSE251872 baseline PBMC only; within-platform z-scoring; module selection by *two-sided* platform-stratified permutation (10,000 permutations, seed 52) at $p < .05$ with leave-one-patient-out sign stability required in all 27 folds; and the preregistration’s own stopping rule: if no module is selected, the arc ends as a training-stage negative and the evaluation cohort is never touched. One implementation attempt failed before producing output (gzip handling of the per-sample archives; disclosed in the redirect log); the corrected script (8420e2e1) produced the single training execution.

6.5 Result: a clean training-stage negative

No module passed. The best two-sided stratified permutation p is .194 (L9, IL2-STAT5); the other eight range from .54 to .92 (results/lc_p52_train_result.json). Under the corrected two-sided rule there is simply no Hallmark-scale signal separating post-infectious ME/CFS from healthy PBMC in this cohort at this n . The evaluation cohort was never opened for this arc. This is the arc’s honest answer, and per the preregistration’s interpretation clause it means: *this* signature strategy - nine fixed modules, platform-stratified, trained on 27 PBMC samples - finds no transportable ME/CFS state signal. It does not establish that Long COVID and ME/CFS are molecularly inseparable; a better-powered cohort (GSE227375, 69 participants) remains as the approved third attempt.

7 External-service annotation landscape

7.1 Purpose and accounting

Forty-four external scientific services were genuinely used and are ledgered with repo-resident evidence (projects/long_covid/EXTERNAL_SERVICES.md): four core arc services (GEO, Enrichr’s Hallmark library, the HGNC complete set, advisory consultations), eight per-gene primary annotation services across the 18 preregistered signature genes (12 L7 panel genes plus the six comparator genes), four set-level services, and twenty-eight further services in hub sweeps whose hubs are the six comparator genes named a priori in the lane. The count is 42 under the program’s conservative recount (three NCBI eutils database rows collapsed to one) and 39 under the harshest recount (all six NCBI-via-eutils rows collapsed); the ledger records all three counts, the rejected candidates, and their failure evidence. These annotations are context for the preregistered gene sets, not independent replication.

7.2 What the landscape actually says

Four annotation results carry scientific weight beyond gate accounting. First, g:Profiler enrichment of the locked 12-gene L7 panel returns, as its top terms, oxoacid, organic-acid, monocarboxylic-acid and pyruvate metabolic processes ($p \approx 10^{-8}$ – 10^{-6}): the panel intersection of the oxphos module is metabolically coherent, not a residual symbol list. Second, the 18-gene signature union forms a connected STRING module (52 edges among 18 nodes) - in contrast to the IC lane’s honest null, where its miRNA hubs had no physical edges. Third, the druggability gradient across the six comparator hubs is real: JAK1 is Tclin with an approved inhibitor class (tofacitinib label retrieved from OpenFDA), CXCL8, BCL2L1, MAP3K8 and STAT3 are Tchem, OSM is Tbio (Pharos), and DGIdb interaction counts vary by two orders of magnitude. Fourth, the literature co-mention counts are honestly sparse: PubMed co-mentions of the hub symbols with Long COVID/PASC range from 11 (CXCL8) and 9 (STAT3) down to 0 (BCL2L1, MAP3K8) - the comparator panel is not a literature-saturated list, and neither is the lane’s signal.

7.3 Independent resonance

Two archive-level findings resonate with the preregistered result without being evidence for it. The PRIDE proteomics archive's top Long COVID project (PXD066724) reports plasma oxidative stress, coagulation activation and glycolytic imbalance - the same metabolic neighbourhood as the L7 down-regulation, in an orthogonal analyte. Open Targets resolves the disease entity cleanly (MONDO_0100233), anchoring the ontology used throughout. Neither is claimed as replication; they are recorded so a future lane can test the metabolic-proteomics axis deliberately.

8 Gate audit, stated exactly

The program's per-disease floors are 120 accession-backed record units, 40 genuinely used external services, a 50-substantive-page disease paper, and a positive benchmark or discovery endpoint. The lane's standing:

Records: met at record level, with honest unit labels. The disease-tagged manifest holds 178 Long COVID record units: 3 GSE series records plus 175 nested GSM sample records (the fourth series, GSE251872's 27 samples plus parent record, is in-lane but not yet folded into the shared manifest count). These are record units: they are not 178 patients, studies, or independent datasets, and the nesting is stated wherever the count is used. The underlying person-level bases are 37 donors (GSE226260 training), 47 subjects (GSE275334 evaluation), and 27 subjects (GSE251872 training).

Services: met. 44 genuinely used external services with repo-resident evidence (42 under the conservative recount; 39 under the harshest, with the recovery path recorded in the ledger). Seven rejected candidates are listed with their failure evidence and are not counted.

Paper: in progress, stated honestly. This manuscript is rendered and measured by the program's substance audit (`scripts/audit_research_body.py`), which counts pages carrying at least 300 non-heading research-body words. Its current reading is printed in the reproducibility appendix and updated at every commit; the 50-page gate is not claimed until the audit says so. The lane's rule, inherited from the program, is substance first: an honest shortfall statement beats padding, and padding is treated as a integrity failure, not a style choice.

Benchmark/discovery: open. The lane claims no independently validated benchmark and no named discovery. The one preregistered positive (E1) is a cross-assay transport of a chronic-state signal; it is not a diagnostic endpoint and the ME/CFS separation it would need for clinical meaning has failed twice under locked protocols.

9 Limitations

Sample sizes are small everywhere. 37 training donors, 47 evaluation subjects (only 9 ME/CFS), 27 second-arc training subjects. The locked $|g| > 0.8$ bar on the ME/CFS contrast is deliberately conservative and the advisory review is right that it makes the arc's information asymmetric: the failures are weak evidence of absence, the passes would have been strong evidence of presence.

The evaluation panel is narrow and chosen by others. GSE275334 measures 635 genes selected for immune phenotyping; mitochondrial and metabolic coverage is incidental. The L7 module survives on 12 panel genes; a metabolic-panel evaluation could look different in either direction.

Cross-assay transport cuts both ways. RNA-seq whole blood (training), RNA-seq PBMC (second training), and NanoString PBMC (evaluation) differ in dynamic range, normalization and cell mixture. The E1 pass shows the signal is robust to that; the negatives may partly be transport artefacts rather than biology, and the manuscript says so rather than resolving it by assertion.

The ME/CFS evaluation arm is small and its own diagnostic purity is unverifiable from GEO metadata;

GSE275334’s healthy arm is healthy, not recovered-from-infection, so E1 effect sizes likely overstate what a recovered-control design would give.

P51’s selection was one-sided in the observed direction. Disclosed, not re-run, corrected in P52. The reader should treat the P51 training p -value as optimistic and the structure of the result (which module, which direction) as the durable content.

The P52 training cohort is platform-split 3/7 and 9/8; stratification protects the permutation null but cannot remove gene-specific platform effects, as the advisory review noted. The leave-one-platform-out sensitivity is computed and reported with the result.

10 Reproducibility appendix

Preregistrations (committed and pushed before any run of their arc): `projects/long_covid/prereg/LC_P51_prereg` (commit 42deadfa); `LC_P52_preregistration.md` (commit f81bb92d, amended pre-freeze from the advisory review as recorded in `redirect/lc_p52_judge_consult.md`).

Analysis scripts and results: `scripts/lc_p51_train.py/results/lc_p51_train_result.json` (commit 858e63f0); `scripts/lc_p51_evaluate.py` (locked at 54b852cc)/`results/lc_p51_eval_result.json` (commit 61d3d6a1); `scripts/lc_p52_train.py` (locked at ff9d5463, pre-result bugfix 8420e2e1)/`results/lc_p52_train_result.json` (commit 889be703). Pre-arc nulls: `results/longcovid_p36_result.json`, p37, p38. Annotation sweeps: `scripts/lc_annotate_*.py`, `projects/long_covid/results/external/`.

Data locks. Every dataset file carries a SHA-256 digest asserted at script start: GSE226260 counts and matrix (p36), GSE275334 normalized panel (p37), GSE334857 raw archive (p38), GSE251872 raw archive and both platform series matrices (p52). Scripts fail closed on mismatch.

Substance audit. The current reading of `scripts/audit_research_body.py` on this PDF is recorded in the commit message of every manuscript revision; at the scaffold commits it read 1 of 50 substantive pages (5 rendered), then 3 of 50 (7 rendered). The count is updated honestly as chapters land; the gate is claimed only when the audit passes.

11 Statistical methods appendix

11.1 Module scoring

For a candidate module M with measurable gene set G_M , each sample’s module composite is the unweighted mean of per-gene z-scores over G_M . Training-cohort expression is library-size normalized to counts per million, filtered (CPM > 1 in at least 20% of samples), and $\log_2(\text{CPM}+1)$ transformed. In P51, z-scores are fit across the training samples; in P52, z-scores are fit *within each sequencing platform* and then pooled, the preregistered guard against the cohort’s platform split. Evaluation z-scores are fit across all 47 evaluation subjects on $\log_2(\text{normalized}+1)$ values, so no group label enters any scaling.

11.2 Effect sizes and permutation tests

Group contrast is Cohen’s D on composites (mean difference); Hedges g adds the small-sample correction. Training selection in P51 used 10,000 label permutations (seed 20260926), one-sided in the observed direction - the disclosed anti-conservatism. P52 used 10,000 *two-sided* permutations with labels permuted *within platform strata* (seed 52), so the null preserves platform membership exactly. Evaluation tests used 10,000 permutations (E1 one-sided in the direction fixed by the untouched training cohort; E2 two-sided). All permutation p -values use the $(b+1)/(B+1)$ convention. Leave-one-patient-out stability requires the sign of D to be unchanged in every fold (37/37 in P51; 27/27 required in P52).

11.3 Eligibility bars, locked a priori

A module is selectable only if at least 20 of its genes are measurable in the training cohort and at least 10 are on the 635-gene evaluation panel. The bars exist because the panel is small and module scores degrade below it; they were fixed in the P51 preregistration before the panel's gene list was consulted for anything but the count.

11.4 Complete module-level results

P51 training (GSE226260, 17 PASC vs 20 recovered; p one-sided observed direction; stable = LOPO sign-stable folds):

module	n genes	D	perm p	stable	selected
L1 _i <i>fna</i>	95	-0.134	0.2548	37/37	no
L2 _i <i>fng</i>	188	-0.164	0.1724	37/37	no
L3 _i <i>nflammatory</i>	153	-0.072	0.2872	37/37	no
L4 _i <i>nfa_nfkf</i>	171	+0.127	0.1721	37/37	no
L5 _i <i>l6_jakstat3</i>	69	-0.174	0.1072	37/37	no
L6 _c <i>omplement</i>	161	-0.196	0.1111	37/37	no
L7 _o <i>xphos</i>	195	-0.356	0.0432	37/37	yes
L8 _h <i>ypoxia</i>	139	-0.130	0.1024	37/37	no
L9 _i <i>l2_stat5</i>	166	-0.106	0.2071	37/37	no

P52 training (GSE251872, 12 PI-ME/CFS vs 15 healthy; p two-sided platform-stratified; no module selected, arc ended at training):

module	n train	n panel	D	perm p	stable	selected
L1 _i <i>fna</i>	95	15	+0.021	0.8917	19/27	no
L2 _i <i>fng</i>	189	52	+0.030	0.7825	26/27	no
L3 _i <i>nflammatory</i>	156	48	-0.028	0.8286	26/27	no
L4 _i <i>nfa_nfkf</i>	179	48	+0.076	0.6851	27/27	no
L5 _i <i>l6_jakstat3</i>	72	35	-0.023	0.8170	24/27	no
L6 _c <i>omplement</i>	163	30	-0.057	0.5444	26/27	no
L7 _o <i>xphos</i>	199	12	-0.014	0.9217	19/27	no
L8 _h <i>ypoxia</i>	145	17	+0.031	0.6303	27/27	no
L9 _i <i>l2_stat5</i>	175	39	+0.074	0.1938	27/27	no

The contrast between the two tables is the lane's methodological lesson in one view: a nominally suggestive one-sided landscape (P51, where several modules sit near .10–.29) becomes a flat null under the corrected two-sided stratified rule (P52), at smaller n with a platform confound absorbed.